

Thibaut Montes

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EDUCATION

PHD IN NUMERICAL PROBABILITY | LPSM | SORBONNE UNIVERSITY

Mar 2017 – Jun 2020

I was under the direction of **Gilles Pagès** and **Vincent Lemaire** at the LPSM and the supervision of **Jean-Michel Fayolle** at ICA.

During my PhD, I made contributions to the theoretical study and financial applications of **Optimal Quantization**, also known as K-means. I also had a keen interest in **Multilevel Monte-Carlo** methods and **Stochastic Algorithms**.

- ◇ Developed numerical methods (fixed-point iteration and gradient descent) for the efficient construction of optimal quantizers.
- ◇ Applied the framework to **PRDC Bermudan** option pricing and to the study of the **stationary Heston** model.

RESEARCH MASTER IN PROBABILITY AND FINANCE (MASTER EL-KAROUI - WITH HONORS) | SORBONNE UNIVERSITY IN COLLABORATION WITH ÉCOLE POLYTECHNIQUE

Sep 2014 – Jun 2016

BACHELOR DEGREE IN MATHEMATICS (WITH HONORS) | AIX-MARSEILLE UNIVERSITY

Sep 2011 – Jun 2014

PROFESSIONAL EXPERIENCE

QUANTITATIVE RESEARCHER | ABU DHABI INVESTMENT AUTHORITY (ADIA)

July 2023 - Now | Abu Dhabi, United Arab Emirates

- ◇ Delivered > 10 signals used in live **Market Neutral** and **Directional** strategies traded daily in US and EU equity markets with Sharpe ratios between 1 and 2.
- ◇ **Co-founded and architected** an NLP/LLM research platform for large-scale extraction of trading signals from unstructured financial text.
- ◇ Thought and implemented **sentence-granular pipelines** for earnings calls, filings, and other unstructured data (parsing, splitting, labelling, embedding) to **generate alpha features** for systematic strategies with emphasis on Point-in-Time data.
- ◇ Designed efficient production and backtest system for systematic LLM and ML/DL (XGBoost/Transformer-based) signals.

SENIOR MACHINE LEARNING RESEARCHER | TALKWALKER

May 2021 – June 2023 | Luxembourg

- ◇ Owned the full lifecycle of NLP models, from **initial research** to **production deployment**, processing up to **10,000 documents per second**.
- ◇ Developed and scaled multilingual **Transformer-based** pipelines for aspect-based sentiment and emotion extraction from raw social media data.
- ◇ Improved classifier accuracy in low-data regimes by implementing **Active Learning** and **Meta-Learning** research.
- ◇ Optimized deep learning inference and data filtering logic to ensure high-fidelity signals across massive, unstructured data streams.

DESK STRAT | DEUTSCHE BANK

Sep 2020 - April 2021 | London, United-Kingdom

I worked on the Synthetic Desk's (ETFs and Indices) migration to Kannon (New P&L and Risk tool used by trading). My main tasks was to improve PV and Risk computations of ETFs and indices.

RESEARCH ADVISOR | THE INDEPENDENT CALCULATION AGENT (NOW OPENSEE)

May 2016 - Mar 2020 | Paris, France

First as an intern, then as a PhD candidate, I worked in collaboration with ICA on the following projects:

- ◇ Optimization of the analytic library using Optimal Quantization (Pricing of Exotic Options in the interest rate world).
- ◇ Identifying and *killing* bias in xVA computation using Multilevel Monte-Carlo methods.

SKILLS

PROGRAMMING

- ◇ **Python** (Pandas, Huggingface, PyTorch, ONNX, NumPy, pytest, Pybind11, ...)
- ◇ **SQL** (Redshift)
- ◇ **C++** (Creation of libraries during my PhD)
- ◇ Tools used often: **Git** ◇ **Airflow** ◇ **S3**
- ◇ Basics knowledge: **Java** ◇ **Scala** ◇ **Kafka** ◇ **Docker**

LANGUAGES

- ◇ **French**: native
- ◇ **English**: fluent

PUBLICATIONS

- ◇ Lemaire, V., Montes, T. and Pagès G. (2022) **Stationary Heston model: Calibration and Pricing of exotics using Product Recursive Quantization**. *Quantitative Finance*, 2022.
- ◇ Fayolle, J.-M., Lemaire, V., Montes, T. and Pagès G. (2021) **Quantization-based Bermudan option pricing in the FX world**. *Journal of Computational Finance*, 2021.
- ◇ Lemaire, V., Montes, T. and Pagès G. (2020) **New weak error bounds and expansions for optimal quantization**. *Journal of Computational and Applied Mathematics*, 2020.